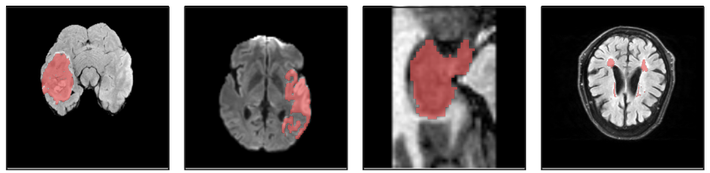
Which adaptation strategy for a new task? What the task inherits decides it: train the large head; fit the small one properly before moving the encoder

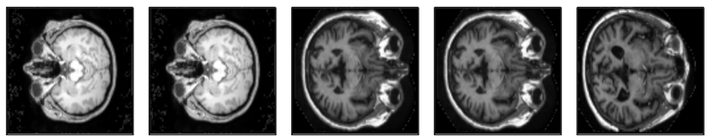
(a) Nine tasks, six cohorts, 32 labelled volumes each

Large task head: inherits the 8.45M-parameter decoder



BraTS glioma | ISLES stroke | Hippocampus | WMH, 5 scanners

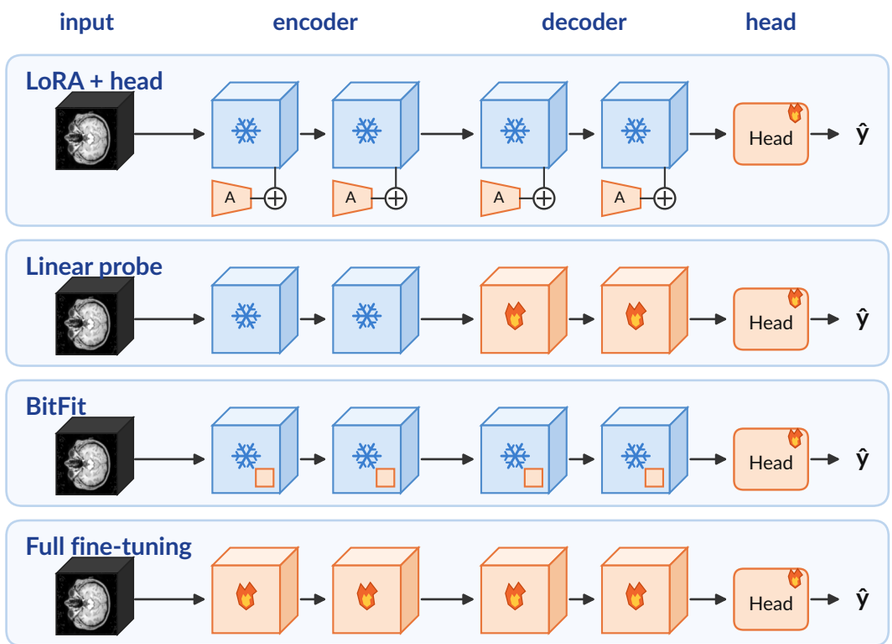
Small task head: ~1K-parameter head on pooled features



IXI age | IXI sex | OASIS age | OASIS sex | OASIS CDR

One held-out test slice per task; segmentation targets in red. Six cohorts, five scanners; the head size is a property of the task that is known before any training run.

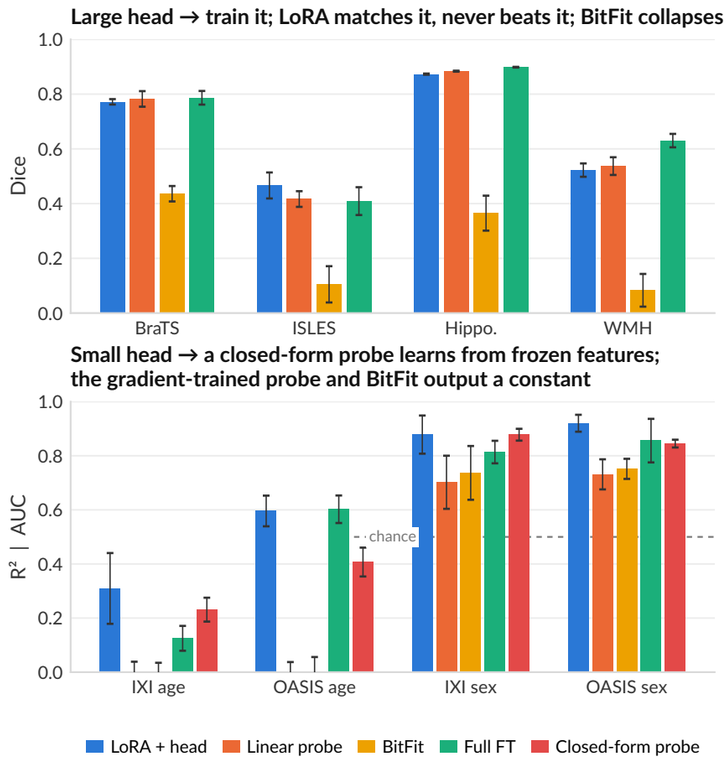
(b) Adaptation strategies: what is frozen and what is trained



Input: 3D MRI, C × 64³ patch (96³ whole head for age / sex). Backbone: FOMO 3D U-Net; 32 volumes, 100 epochs, one recipe.

= frozen | = trained for the new task | A = low-rank adapter (LoRA) | = trainable bias (BitFit)

(c)



Forgetting: a second segmentation task erases the first in either order (BraTS 0.78→0.20); EWC softens it, only freezing prevents it.

Pretraining: 59 of 82 tensors load; under LoRA it equals random init on four tasks (+0.000 Dice on BraTS); frozen features beat random.

Protocol: subject-grouped splits, whole-volume inference, 5 seeds, 676 runs, seed-level paired tests, per-strategy learning rate.